CLASS 9 MATHEMATICS • NCERT 0904CH03

The World of Numbers

An exhaustive, point-by-point classroom presentation charting humanity's voyage from prehistoric tally bones to Vedic astronomical powers, Brahmagupta's laws of zero and negative debts, rational density, the crisis of irrationals, cyclic numbers, and the continuous real line.

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The Human Need to Count: One-to-One Correspondence

The Cattle & Pebble Model

• Long before number symbols or words existed, humans solved survival needs using **one-to-one correspondence**.

• **Morning:** For every cow leaving to graze in the Saraswati river valley forests, place one pebble in a clay pot.

• **Evening:** For every returning cow, remove one pebble from the pot.

• *Empty pot* ⟹ Herd complete; *Pebbles remaining* ⟹ Cattle missing!

Birth of Natural Numbers (ℕ)

• Matching physical objects to tallies gave rise to the **Natural Numbers**:

$$\mathbb{N} = \{1, 2, 3, 4, 5, \ldots\}$$

THE DISCRETE COUNTING SET

• **Closure Property:** Natural numbers are closed under addition (sum of any two natural numbers is always a natural number), but *not* closed under subtraction (e.g. $3 - 5 \notin \mathbb{N}$).

The Dawn of Number Sense

Mathematics did not begin in lecture halls with abstract equations; it began in the soil, on the bark of trees, on finger joints, and notched into animal bones to track assets and time.

Pages 41-42 (Fig 3.1)

Mathematical Artifacts: Lebombo & Ishango Bones

Lebombo Bone (~35,000 BCE)

- Discovered in the Lebombo Mountains between South Africa and Eswatini.
- A baboon fibula featuring **29 deliberate, uniformly carved notches**.
- Used as a **lunar phase counter** or menstrual calendar.
- First indisputable physical evidence of humans recording numbers to measure periodic cycles.

Ishango Bone (~20,000 BCE)

- Discovered near the headwaters of the Nile River in DR Congo.
- Features three columns of asymmetrical tally groupings.
- **Prime Column:** Groups notches into **11, 13, 17, and 19** (the exact primes between 10 and 20!).
- **Doubling Column:** Demonstrates multiplication by 2 (3 to 6, 4 to 8). Proof of advanced abstract number sense!

Tactile Counting: The Finger Joint System

Traditional Indian finger counting uses the thumb to count the 3 distinct joints on each of the 4 fingers. This yields **12 counts per hand**, forming the historical foundation for ancient base-12 arithmetic systems.

Indus Trade & Vedic Decimal Nomenclature

Trade at Lothal & Harappa

• Ancient urban dockyards and trading hubs required standardized weights and measures.

• Merchants exchanging cotton, lapis lazuli, and spices with Mesopotamia developed strict tallying systems.

Vedic Powers of Ten

• In Vedic literature, individual names were assigned to each power of 10 up to **10^{12} (parārdha)**.

• The *Rigveda* explicitly structured mathematical verses around powers of ten, laying the base for the modern decimal system.

Lalitavistara (~4th c. BCE)

• The Buddha recites numerical scales going far beyond worldly needs, reaching **10^{53} (tallakṣaṇa)**.

• Shows deep philosophical and scientific comfort with cosmic, astronomical magnitudes.

Decimal Place-Value Foundation

This unbroken cultural mastery of powers of 10 and positional notation in India provided the essential fertile ground for the greatest mathematical invention in human history: the operational **Zero (Śhūnya)**.

Check Your Understanding: Prehistoric & Vedic Math

Q1 How does the cattle-and-pebble method demonstrate the mathematical principle of "one-to-one correspondence"?

Q2 What special mathematical property links the numbers 11, 13, 17, and 19 carved into the 20,000-year-old Ishango bone?

Q3 Why is the set of Natural Numbers ℕ closed under addition, but not closed under the operation of subtraction?

Q4 How did counting the 3 joints of each finger with the thumb lead ancient cultures to develop base-12 counting systems?

The Revolution of Shūnya: When Nothing Became Something

The Ancient World's Dilemma

- For millennia, counting began strictly at 1. Five apples given away meant an empty void, not an operable number.
- Babylonians and Mayans created *placeholders* (empty punctuation marks to distinguish 305 from 35).
- However, they never performed arithmetic with the void: you could not add, subtract, or multiply by a placeholder!

From Śhūnyatā to Mathematics

- In the Upanishads, Buddhist texts, and Patañjali's *Yoga Sūtras* (c. 3rd c. BCE), **Śhūnyatā** (emptiness) represented the serene state of stilling all mental fluctuations.
- Because Indian thinkers revered emptiness philosophically, they possessed the conceptual readiness to embrace "nothingness" as a tangible mathematical reality.

Philosophical Stillness to Mathematical Power

The philosophical concept of emptiness crystallised into the mathematical zero through Āryabhaṭa and reached full operational algebraic maturity in the work of Brahmagupta (628 CE).

The Bakhshālī Manuscript & Brahmagupta's Laws

The Bakhshālī Bindu (Early CE)

• Written on birch bark, the Bakhshālī Manuscript preserves the historic transition from a space to an explicit dot: the **bindu (•)** representing zero.

• But a symbol is just ink until arithmetic rules govern it. That breakthrough belongs to **Brahmagupta** in the *Brāhmasphuṭasiddhānta* (628 CE).

Brahmagupta's Definition of Zero

Brahmagupta explicitly defined zero as the mathematical result of subtracting a number from itself:

$$a - a = 0$$

ZERO AS AN ALGEBRAIC IDENTITY

Zero was now an active, functioning citizen of the number world!

Brahmagupta's Three Fundamental Laws for Zero

1. **Addition:** When zero is added to a number, it remains unchanged: **a + 0 = a**.
2. **Subtraction:** When zero is subtracted from a number, it remains unchanged: **a − 0 = a**.
3. **Multiplication:** When any number is multiplied by zero, the product is zero: **a × 0 = 0**.

Check Your Understanding: The Birth of Zero

Q1 What is the crucial difference between a "placeholder symbol" (like in Babylon) and Brahmagupta's operational zero?

Q2 How does the Indian philosophical concept of Śhūnyatā relate to the birth of zero in Indian mathematics?

Q3 What physical mark was used in the ancient Bakhshālī Manuscript to visually denote zero?

Q4 State Brahmagupta's definition of zero in terms of subtraction, as recorded in the Brāhmasphuṭasiddhānta (628 CE).

Integers: Fortunes (Dhana) and Debts (Rina)

Fortunes (Dhana) = Positive Integers

- Represent assets, credits, or wealth owned.
- Numbers to the right of zero on the number line: +1, +2, +3, +4...
- Gain in elevation, temperature above zero, money in hand.

Debts (Rina) = Negative Integers

- What happens when you subtract 5 from 3? (3 - 5 = -2).
- Brahmagupta grounded negative numbers in real commercial debt: owing money.
- Numbers to the left of zero on the number line: -1, -2, -3, -4...

$$\mathbb{Z} = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$$

THE SET OF INTEGERS (FROM GERMAN 'ZAHLEN', MEANING NUMBERS)

Completing the Number Line Axis

By defining zero as the central origin and moving left into the realm of debts, Brahmagupta extended the one-dimensional number line indefinitely in both directions.

The Arithmetic Laws of Fortunes & Debts

+ Addition Rules

• **Fortune + Fortune = Fortune:**
5 + 4 = 9

• **Debt + Debt = Debt:**
(-5) + (-4) = -9
(Owing ₹5 and borrowing ₹4 means owing ₹9)

− Subtraction Rules

• **Fortune - 0 = Fortune:**
7 - 0 = 7

• **Debt - 0 = Debt:**
-6 - 0 = -6

• **Subtracting a Debt:**
10 - (-5) = 15

× Multiplication Rules

• **Debt × Fortune = Debt:**
(-3) × 4 = -12

• **Debt × Debt = Fortune:**
(-3) × (-4) = +12

The product of two debts is always a fortune!

Think & Reflect: Why Does (−) × (−) = (+)?

Think in terms of action on debt: A negative number is a debt. Multiplying by a negative number means **taking away** that debt. If a creditor cancels four debts of ₹3 each (-4 × -3), you are effectively **₹12 richer (+12)**!

Check Your Understanding: Signed Numbers

Q1 If the temperature in Ladakh is 4 °C at noon and drops by 15 °C by midnight, what integer represents the midnight temperature?

Q2 A merchant has a debt of ₹850, earns a profit of ₹1,200, and then incurs a loss of ₹450. What is his net financial standing?

Q3 Calculate the following using Brahmagupta's laws: (i) (−12) × 5, (ii) (−8) × (−7), and (iii) 0 − (−14).

Q4 Explain using the real-world idea of debt cancellation why 10 − (−5) equals 15.

Filling the Spaces: The Set of Rational Numbers (ℚ)

The Need for Parts of a Whole

• Counting herds required integers, but measuring land, dividing crops, or half-cups of ghee required **fractions**.

• Just as every integer has an additive inverse (3 has -3), positive fractions have **negative fractions** (−¾ for ¾).

• Note on negative signs: **−(1/5) = (−1)/5 = 1/(−5)**.

÷ Definition of a Rational Number (ℚ)

Any number that can be expressed in the form:

$$p / q \text{ , where } p, q \in \mathbb{Z} \text{ and } q \neq 0$$

ℚ = SET OF RATIONAL NUMBERS (FROM 'QUOTIENT')

• All integers are rational (5 = 5/1, −10 = −10/1).

• Standard form requires p and q to be **co-prime**.

Why Must q ≠ 0?

Division by zero is undefined because no unique number x exists such that 0 × x = p (for non-zero p), and for p = 0, every number satisfies it, destroying all algebraic consistency.

Brahmagupta's Laws for Rational Arithmetic

= Equality

$a/b = c/d \iff ad = bc$

Cross-multiplication test for equivalence.

± Add / Subtract

$a/b \pm c/b = (a \pm c)/b$

Convert to common denominator b first.

× Multiply

$(a/b) \times (c/d) = (ac)/(bd)$

Direct product of numerators and denominators.

÷ Division

$(a/b) \div (c/d) = (ad)/(bc)$

Multiply by reciprocal ($c \neq 0$).

Closure, Commutativity, and Distributivity

- **Closure:** $\mathbb{Q}$ is closed under addition, subtraction, and multiplication. It is also closed under division (except dividing by zero).
- **Commutativity & Distributivity:** $a/b + c/d = c/d + a/b$, and $p(q + r) = pq + pr$ for all rationals p, q, r.

Check Your Understanding: Rational Operations

Q1 Using the cross-multiplication rule ad = bc, check whether 5/4 and 10/8 represent the exact same rational number.

Q2 Why must we find a common denominator before adding two rational numbers like 2/5 and 3/10?

Q3 What is the result of dividing (−4/7) by (5/14)? (Hint: multiply by the reciprocal).

Q4 Is the set of rational numbers $\mathbb{Q}$ closed under division by all rational numbers? What single exception exists?

Representation on the Number Line & Absolute Value

Plotting p/q on the Line

- To plot **p/q (q > 0)**, divide each unit interval into **q equal parts**.
- Starting from 0, count **p parts** to the right (if positive) or left (if negative).
- *Example:* 3/4 divides [0, 1] into 4 parts; take 3 parts. 9/4 = 2 + ¼, so it lies one-fourth past 2.

Absolute Value & Distance

- **Absolute Value |x|:** Distance of x from 0 on the number line. Always non-negative: **|x| ≥ 0**.
  - |5/3| = 5/3 and |-5/3| = 5/3.
- **Distance Between Two Rationals a and b:** Given by **|a − b|**.
  - Distance between -4 and 3: |3 − (−4)| = **7 units**.

Geometrical Invariance of Distance

Because |a − b| = |b − a|, distance on the number line is independent of direction. Distance is purely a measure of spatial separation.

The Magical Density of Rational Numbers

> **The Density Theorem for Rationals**
> Between **any two distinct rational numbers**, no matter how close, there are **infinitely many** other rational numbers!

The Midpoint / Average Method

For any two rational numbers a and b with a < b:

$$m = (a + b) / 2$$

GUARANTEED MIDPOINT: A < M < B

- Between 1 and 2 ⟹ (1 + 2)/2 = **3/2**.
- Between 1 and 3/2 ⟹ (1 + 3/2)/2 = **5/4**.

The Illusion of Completeness

- Because rational numbers are dense, it feels as though they must pack the entire number line with zero gaps.
- Yet, ancient builders and philosophers discovered that the line has **unfillable holes** that no fraction can ever touch!

Check Your Understanding: Number Line & Density

Q1 Between which two consecutive integers does the improper fraction −5/4 lie on the number line?

Q2 What is the absolute distance between the rational numbers −1/2 and 1/4 on the number line?

Q3 Why is the average (a + b)/2 of two distinct rational numbers a and b guaranteed to be a rational number lying strictly between them?

Q4 If there are infinitely many rational numbers between 3.1415 and 3.1416, name two such rational numbers.

Irrational Numbers: The Diagonal of the Unit Square

Baudhāyana's Śhulbasūtra (c. 800 BCE)

- Manual for constructing ritual Vedic fire altars (Vedi).
- For a square altar with side 1 unit, the diagonal d satisfies:
  $d^2 = 1^2 + 1^2 = 2 \Longrightarrow d = \sqrt{2}$.
- Baudhāyana discovered that this length could not be matched by any ratio of integers, requiring ingenious fractional approximations.

What Is an Irrational Number?

- A real number that **CANNOT** be expressed as a ratio p/q of integers ($q \neq 0$).
- Their existence proved that the rational numbers, despite being dense, leave countless **holes** on the number line!
- Every square root of a non-perfect square ($\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}$) is strictly irrational.

The Historical Shockwave

Ancient Greek Pythagoreans believed all cosmic harmony was expressible as whole-number ratios. Discovering that the diagonal of a simple 1×1 square defied all fractions shook the ancient mathematical world to its core!

Hippasus' Classical Proof: Why √2 Is Irrational

Logical Steps of the Proof

1. **Assumption:** Assume √2 is rational, so **√2 = p/q** in simplest form (p, q co-prime, q ≠ 0).
2. **Square both sides:** 2 = p² / q² ⟹ **p² = 2q²**.
3. **Deduction for p:** p² is even, which forces **p to be even**. Let p = 2k.
4. **Substitute back:** (2k)² = 2q² ⟹ 4k² = 2q² ⟹ **q² = 2k²**.
5. **Deduction for q:** q² is even, which forces **q to be even**!

The Fatal Contradiction

> **Contradiction:** Both p and q are even, meaning they share a common factor of 2. But we assumed p/q was in simplest form with NO common factor!

• Since every algebraic step was mathematically sound, our initial assumption must be false.

• **Conclusion:** √2 cannot be written as a fraction. **√2 is irrational!** ✅

Check Your Understanding: The Logic of Contradiction

Q1 What fundamental strategy defines a "Proof by Contradiction"?

Q2 In the proof of $\sqrt{2}$, why does $p^2 = 2q^2$ guarantee that p must be an even integer?

Q3 Where does the logical contradiction occur after showing that both p and q are even?

Q4 Can the exact same proof technique be extended to prove that $\sqrt{3}$ and $\sqrt{5}$ are irrational numbers?

Constructing Length √n & The Square Root Spiral

Locating √2 with Ruler & Compass

- **Step 1:** On the number line, mark segment OA = 1 unit from Origin O(0).
- **Step 2:** Erect perpendicular AB = 1 unit at A. Join OB.
  By Baudhāyana's Theorem: **OB = √(1² + 1²) = √2**.
- **Step 3:** With center O and radius OB, draw a circular arc cutting the number line at P.
- Point P represents the exact location of **√2 (≈ 1.414...)**!

The Square Root Spiral (Theodorus)

- Erect a perpendicular of 1 unit onto hypotenuse OB(√2) ⟹ new hypotenuse = **√((√2)² + 1²) = √3**.
- Erect 1 unit onto √3 ⟹ hypotenuse = **√4 = 2**.
- Continue step-by-step to generate **√5, √6, √7... √n**.
- Forms an elegant mathematical spiral of successive right-angled triangles!

Geometry Validates What Arithmetic Doubted

Even though √2 has a never-ending decimal expansion that cannot be written as a fraction, its physical length exists with absolute precision on the number line!

The Story of Pi (π) & Mādhava's Infinite Series

Āryabhaṭa (499 CE)

• Provided the stunning approximation: **62832 / 20000 = 3.1416**.

• He explicitly noted this was only *āsanna* (an approximation), intuiting that π could never be captured by a finite fraction!

Mādhava of Sangamagrama

• Founder of the 14th-century Kerala School of Mathematics.

• Realized that an irrational number cannot be computed by a single fraction, discovering the world's first **infinite series**:

$$\pi = 4 \times (1 - \tfrac{1}{3} + \tfrac{1}{5} - \tfrac{1}{7} + \ldots)$$

Lambert's Proof (1761)

• Swiss mathematician Johann Lambert formally proved what Āryabhaṭa intuited 1,260 years earlier: **π is strictly irrational**.

• π cannot be written as any fraction p/q.

What Does It Mean to Add Infinite Terms?

Adding infinitely many terms means examining the value that the partial sums get progressively closer and closer to. This breakthrough by Mādhava marked the birth of mathematical analysis and calculus.

Check Your Understanding: Spiral & Infinite Series

Q1 How does erecting a 1-unit perpendicular on a line of length √2 help us construct a line segment of length √3?

Q2 Why did Āryabhaṭa describe his approximation 3.1416 for π as an "āsanna" (approximated) value?

Q3 What was the revolutionary insight made by Mādhava of Sangamagrama in expressing the value of π?

Q4 Is the familiar number 22/7 the exact value of π, or merely a convenient rational approximation?

Rational Decimals: Terminating vs. Repeating

Terminating Decimals

• Division leaves a remainder of 0; decimal stops.

• *Example:* 3/8 = 0.375; 7/20 = 0.35.

• **The Golden Rule:** The decimal of p/q (in lowest terms) terminates **if and only if** the prime factorisation of the denominator q contains **only 2s, only 5s, or both 2 and 5** (i.e. $q = 2^a \cdot 5^b$)!

Non-Terminating Repeating Decimals

• Division never leaves remainder 0; digits loop forever.

• *Example:* 1/3 = 0.333... = $0.\overline{3}$; 5/11 = 0.4545... = $0.\overline{45}$

• **Why must they repeat?** When dividing by q, only (q − 1) non-zero remainders are possible. By the Pigeonhole Principle, a remainder must recur, forcing the quotient to loop!

Predicting Decimals Without Long Division

Consider 3/20: Since $20 = 2^2 \times 5$, multiply top and bottom by 5 to make the denominator 100 $\Longrightarrow$ 15/100 = 0.15. If any prime factor other than 2 or 5 appears in lowest terms (e.g. 3, 7, 11), it *must* be repeating!

Converting Repeating Decimals into p/q Form

Pure Repeating: 0.45^{-}

• Let x = 0.454545...

• Two digits repeat, so multiply by $10^2 = 100$:
100x = 45.454545...

• Subtract: 100x − x = $45.45^{-} - 0.45^{-}$
99x = 45 ⟹ **x = 45/99 = 5/11** ✅

$0.45^{-} = 5 / 11$

General Repeating: 0.16^{-}

• Let x = 0.16666...

• Shift 1 non-repeating digit: **10x = 1.666...**

• Shift 1 repeating cycle: **100x = 16.666...**

• Subtract: 100x − 10x = $16.6^{-} - 1.6^{-}$
90x = 15 ⟹ **x = 15/90 = 1/6** ✅

$0.16^{-} = 1 / 6$

Universal Rule for Conversion

Multiply by 10^m (m = non-repeating digits), then by 10^n (n = repeating block length). Subtracting perfectly eliminates the infinite recurring tail!

Check Your Understanding: Decimals & Fractions

Q1 Without dividing, does the fraction 13/250 terminate or repeat? How many decimal places will it have?

Q2 Why will the rational fraction 4/15 always produce a repeating decimal rather than a terminating one?

Q3 Convert the pure repeating decimal $0.\overline{6}$ into its simplest fraction form p/q.

Q4 What power of 10 must you multiply x by to clear the non-repeating part in the number x = 2.357?

The Magic of Cyclic Numbers & 0.999... = 1

The Magic of 1/7 = $0.\overline{142857}$

Multiplying the 6-digit repeating block by 1 through 6 preserves the exact cyclic sequence of digits:

```
142857 × 1 = 142857
142857 × 2 = 285714
142857 × 3 = 428571
142857 × 4 = 571428
142857 × 5 = 714285
142857 × 6 = 857142
```

The Mystery: Why $0.\overline{9} = 1$ Exactly!

- Let x = 0.99999...
- Multiply by 10: **10x = 9.99999...**
- Subtract: 10x − x = $9.\overline{9} - 0.\overline{9}$
- **9x = 9 ⟹ x = 1**!

0.999... is NOT "almost 1"; it is identically 1!

Non-Uniqueness of Decimal Representations

Every terminating decimal has an alternate representation with trailing repeating 9s (e.g. 1.000... = 0.999... and 2.47000... = 2.46999...). Rational decimals are dual-faced!

Check Your Understanding: Cyclic Numbers & $0.\overline{9}$

Q1 What unique cyclical property occurs to the digits 142857 when you evaluate 2/7, 3/7, and 4/7?

Q2 Using the algebraic subtraction method, prove why 0.999... is mathematically identical to 1.

Q3 How can the number 3.5000... be alternatively represented as an infinite repeating decimal with 9s?

Q4 Why does the decimal expansion 1.01001000100001... represent an irrational number, even though it follows an obvious rule?

The Real Number System & Beyond the Line

Nested Hierarchy of Numbers

- **Natural Numbers (ℕ):** Counting set {1, 2, 3...}.
- **Integers (ℤ):** Adding zero and negative debts {...-2, -1, 0, 1, 2...}.
- **Rational Numbers (ℚ):** All fractions p/q (terminating or repeating decimals).
- **Irrational Numbers (𝕀):** Never-ending, non-repeating (√2, π, √10).
- **Real Numbers (ℝ) = ℚ ∪ 𝕀 (The complete, unbroken line!)**

The Final Frontier: Stepping Off the Line

What is the square root of -1?

- 1 × 1 = 1, and (−1) × (−1) = 1. No real number squared can ever yield -1!
- Mathematicians stepped completely off the 1D line into a 2nd dimension, defining:

$$i = \sqrt{(-1)}$$

IMAGINARY UNIT • COMPLEX PLANE

Powers modern quantum physics, electronics, and smartphones!

Check Your Understanding: The Number Realm

Q1 Is every integer a rational number? Is every rational number an integer? Justify with counter-examples.

Q2 Can a number be both rational and irrational at the same time?

Q3 What property of the Real Number line distinguishes it from the set of rational numbers alone?

Q4 Why cannot the square root of -1 exist anywhere along the continuous Real Number line?

Representative Worked Problems from the Chapter

Convert 3.052 ̄to p/q

Problem: Convert x = 3.05252... into p/q.

Solution:

• 10x = 30.5252... (Eq 1)

• 1000x = 3052.5252... (Eq 2)

• Subtract Eq 1 from Eq 2:

990x = 3022

x = 3022 / 990 = 1511 / 495

Ans: 1511 / 495 ✅

Decimal Places of 18/125

Problem: Predict decimal type of 18/125 without division.

Solution:

• 125 = 5^3.

• Contains only 5s ⟹ **Terminating**!

• $18/5^3 = (18 \times 2^3) / (5^3 \times 2^3) = (18 \times 8) / 1000 =$ **144 / 1000 = 0.144**.

Terminates in 3 places ✅

Algebraic Proof (Ex 14)

Problem: x + y + z = 0 and xy + yz + zx = 0. Prove x = y = z = 0.

Solution:

$(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$

$0^2 = x^2 + y^2 + z^2 + 2(0)$

⟹ **$x^2 + y^2 + z^2 = 0$**.

Squares are non-negative, so x = y = z = 0!

Proved: x = y = z = 0 ✅

Complete Concept Checklist & Core Takeaways

Counting, Zero, Integers & Rationals

- ✅ **Natural Numbers (ℕ):** Began with 1-to-1 correspondence (Ishango/Lebombo tallies).
- ✅ **Śhūnya (Zero):** Defined by Brahmagupta as a − a = 0; transformed emptiness into a number.
- ✅ **Integers (ℤ):** Negative debts (rina) and positive fortunes (dhana); (−) × (−) = (+).
- ✅ **Rational Numbers (ℚ):** Ratios p/q (q ≠ 0) in lowest terms. Closed under +, −, ×, and ÷ (except 0).
- ✅ **Density:** Infinitely many rational numbers between any two points (via average (a+b)/2).

Irrationals, Decimals & The Real Line

- ✅ **Irrational Numbers (𝕀):** Numbers like √2 and π that cannot be written as fractions.
- ✅ **Hippasus' Proof:** Rigorous proof by contradiction establishing that √2 is irrational.
- ✅ **Decimals:** Rationals are terminating ($q = 2^a \cdot 5^b$) or repeating; Irrationals never terminate or repeat.
- ✅ **Cyclic Numbers:** $1/7 = 0.\overline{142857}$ preserves cyclic digit permutations when multiplied by 1–6.
- ✅ **Real Numbers (ℝ):** Unites ℚ and 𝕀 into an unbroken, continuous geometric line.

END OF CHAPTER 3

Numbers are the Music of the Mind, And the Universe is Its Instrument.

"From the first notch on bone to the infinite horizon of the real line, humanity found in numbers the mirror of reality itself."

✦ Akash Srivastva (www.akashmaths.online)